Lecture 3: detailed solutions

Real book exercises — Economics of Organizations and Incentives (MPE Insper)

Solutions to `Listas/lista-reais-lecture-03.pdf`

November 2026

These are detailed solutions to the real book exercises of the list `lista-reais-lectures-01-05.pdf` in `Listas/` (one block of ten per lecture). Within each lecture the items run from easy to hard, in the list's order, and each heading shows the item's number in the lecture, its number in the combined list, its level and its difficulty from 1 (apply a definition) to 5 (graduate level). Statements are reproduced verbatim in the list and are not repeated here. Every solution has the same six parts: (a) what is asked, in one line; (b) the method, with the book section and page and the lecture slide; (c) the solution, every step shown (for matrices, the full dominance comparisons or best-response marking); (d) the economic interpretation; (e) the verification, by the Python check scripts in this folder and against the book's worked answer or the MWG solutions manual where one exists; (f) common mistakes. Books: Tadelis (2013), Muñoz-Garcia & Toro-Gonzalez (2019, 2nd ed.), Gibbons (1992), Mas-Colell, Whinston & Green (1995) (MWG) and Cabral (2017). Notation follows the slides: a game is $G = (I, \{S_i\}_{i\in I}, \{u_i\}_{i\in I})$, s_{-i} is a profile of the others' strategies, S_i^k is the set of player i's strategies that survive k rounds of IESDS with every strictly dominated strategy removed in each round (Lecture 1, slide 29), and $\sigma_i \in \Delta(S_i)$ is a mixed strategy. In bimatrices player 1 picks rows and the first payoff is player 1's; underlined payoffs are best replies.

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Lecture 3 — Dynamic games: extensive form, backward induction, SPE

Item 1 (item 21 of the combined list)**: Tadelis Exercise 7.1** Foundations · difficulty 1/5

Source: Tadelis Exercise 7.1, page 147. The statement is reproduced verbatim in the list (item 21); it is not repeated here.

(a) What is asked. Count player i's pure strategies when he has K information sets with (a) m actions at each, (b) m_k actions at information set k.

(b) Method. A pure strategy is a complete plan: one action at *every* information set, including those the plan itself never reaches (Lecture 3, slide 25; Tadelis §7.2.1, Def. 7.4, pp. 137–138). So the set of pure strategies is the Cartesian product of the action sets, $S_i = \times_{h_i \in H_i} A_i(h_i)$, and its size is the product of their sizes (counting principle).

(c) Solution. **(a)** Label player i's information sets $h^1, \ldots, h^K$. A pure strategy is a list $s_i = (s_i(h^1), \ldots, s_i(h^K))$ with $s_i(h^k) \in A_i(h^k)$ and $|A_i(h^k)| = m$. The choice at h^1 can be made in m ways; for each of them the choice at h^2 in m ways, and so on:

$$|S_i| = \underbrace{m \cdot m \cdots m}_{K \text{ times}} = m^K.$$

(b) The same argument with $|A_i(h^k)| = m_k$:

$$|S_i| = m_1 \cdot m_2 \cdots m_K = \prod_{k=1}^{K} m_k.$$

Examples. In the sequential battle of the sexes (Lecture 3, slide 26) the follower has $K = 2$ information sets with $m = 2$ actions, so $2^2 = 4$ strategies (OO, OF, FO, FF). A player with information sets of sizes $(2, 3, 4)$ has $2 \cdot 3 \cdot 4 = 24$ pure strategies.

Mixed versus behavioral strategies. The count also measures how large the two kinds of randomization are (Lecture 3, slide 27; Tadelis §7.2.2, Defs. 7.5–7.6, pp. 139–140). A mixed strategy is a point in the simplex over $\prod_k m_k$ pure strategies, which has dimension $\prod_k m_k - 1$; a behavioral strategy picks one distribution per information set, dimension $\sum_k (m_k - 1)$. With $(m_k) = (2, 3, 4)$ these are 23 and 6. Under perfect recall the smaller space already generates every distribution over outcomes (Kuhn's theorem, Lecture 3, slide 28; Tadelis p. 142).

(d) Interpretation. The count grows exponentially in the number of information sets: what makes a dynamic game "big" is the number of contingencies, not the number of moves actually played. This is why the normal form of even a small tree (the Brexit game of Lecture 4, slides 7–8, gives the EU nine strategies) is cumbersome, and why backward induction, which works node by node, is the practical tool.

(e) Verification. `check_B_L03.py` enumerates every plan with `itertools.product` for $(m_k) = (2, 2), (3, 2, 2), (2, 3, 4), (5)$ and for $m = 3$, $K = 4$, and compares the count with the formulas; it also checks the two dimensions 23 and 6. Output line: `tadelis-7-1: m^K and prod m_k confirmed by enumeration`. Tadelis prints no answer.

(f) Common mistakes.

- Adding instead of multiplying (Km or $\sum_k m_k$): that counts *actions*, which is the dimension count of behavioral strategies plus K, not the number of plans.
- Leaving out information sets that a plan makes unreachable. A strategy specifies an action there too (the off-path parts are exactly what subgame perfection tests).
- Counting decision nodes instead of information sets: nodes in the same information set share one action choice.

Answer. (a) m^K. (b) $\prod_{k=1}^{K} m_k$.

Item 2 (item 22 of the combined list)**: Muñoz-Garcia Ch. 4, Exercise 3** Foundations · difficulty 1/5

Source: Muñoz-Garcia Ch. 4, Exercise 3, page 160. The statement is reproduced verbatim in the list (item 22); it is not repeated here.

(a) What is asked. In the sequential prisoner's dilemma of Fig. 4.9: (a) player 2's best reply to each move of player 1; (b) player 1's move and the subgame-perfect equilibrium; (c) whether play differs from the simultaneous version (Ch. 1, Fig. 1.8).

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Prop. 8.1, p. 153); in such games the SPE are exactly the backward-induction profiles (Lecture 3, slide 35). An SPE is written as complete strategies: player 2's strategy names a reply at *both* of her nodes, including the one off the path (Lecture 3, slide 25; Tadelis Def. 7.4, p. 138; Gibbons pp. 125–126 on outcome vs. equilibrium). All Nash equilibria of the normal form are found with the best-reply and indifference conditions (principle of indifference, Lecture 3, slides 10–12).

(c) Solution. Notation. C = Confess, N = Not confess. Payoffs (u_1, u_2) from Fig. 4.9: $(C,C) \mapsto (-5,-5)$, $(C,N) \mapsto (0,-15)$, $(N,C) \mapsto (-15,0)$, $(N,N) \mapsto (-1,-1)$. Player 2's strategy is a pair $s_2 = s_2(C)\, s_2(N)$: her reply after C, then after N. So $S_2 = \{CC, CN, NC, NN\}$ and $S_1 = \{C, N\}$.

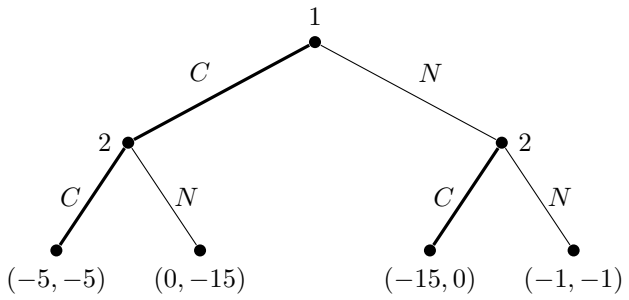


Fig. 4.9 redrawn; thick branches = backward-induction choices.

(a) Player 2's best replies.

$$\begin{aligned} &\text{after } C: \quad u_2(C,C) = -5 > -15 = u_2(C,N) \;\Rightarrow\; C; \\ &\text{after } N: \quad u_2(N,C) = 0 > -1 = u_2(N,N) \;\Rightarrow\; C. \end{aligned}$$

So $s_2^* = CC$: confess whatever player 1 did.

(b) Player 1's move and the SPE. Replacing each of player 2's nodes by its outcome,

$$u_1(C, s_2^*) = u_1(C,C) = -5 \;>\; -15 = u_1(N,C) = u_1(N, s_2^*),$$

so player 1 confesses. The SPE is

$$(s_1^*, s_2^*) = (C,\ (C \text{ after } C,\ C \text{ after } N)) = (C, CC),$$

with outcome (C,C) and payoffs $(-5,-5)$. It is unique: every comparison above is strict.

Normal form and all Nash equilibria (deepening).

	CC	CN	NC	NN
C	$\underline{-5},\ \underline{-5}$	$-5,\ \underline{-5}$	$\underline{0},\ -15$	$\underline{0},\ -15$
N	$-15,\ \underline{0}$	$\underline{-1},\ -1$	$-15,\ \underline{0}$	$-1,\ -1$

Underlined payoffs are best replies. *Pure:* only (C, CC) has both payoffs underlined. *Mixed:* let $p = \Pr_1(C)$.

- If $0 < p < 1$, player 2's payoffs are $-5p$ (CC), $-5p-(1-p)$ (CN), $-15p$ (NC), $-15p-(1-p)$ (NN); CC is the unique best reply, and against CC player 1 strictly prefers C $(-5 > -15)$, so $p = 1$. Contradiction.
- If $p = 0$, player 2's best replies are CC and NC (both reply C to N), and against either one C pays player 1 -5 or 0, more than -15. Contradiction.
- If $p = 1$, player 2's best replies are CC and CN (both reply C to C; they differ only at the unreached node). Let $w = \Pr_2(CN)$. Player 1 keeps C iff

$$u_1(C) = -5 \ \geq\ u_1(N) = -15(1-w) + (-1)w = -15 + 14w \iff w \leq \tfrac{5}{7}.$$

So the Nash equilibria are $\{(C,\ (1-w)\,CC + w\,CN) : 0 \leq w \leq \frac{5}{7}\}$, all with outcome (C, C). Only $w = 0$ is subgame perfect: for $w > 0$ player 2 plans N after N with positive probability, which is not a best reply in that subgame $(-1 < 0)$. In behavioral terms: player 2 plays C after C and, after N, plays N with probability at most $\frac{5}{7}$.

(c) Comparison. In the simultaneous game (Fig. 1.8) C is strictly dominant for each prisoner $(-5 > -15$ and $0 > -1)$, so the unique Nash equilibrium is (C, C) with $(-5, -5)$. The sequential version has the same outcome. Seeing player 1's move does not change player 2's choice, because C is her best reply to each of player 1's moves; player 1 therefore gains nothing from moving first.

(d) Interpretation. Observing the other's move matters only when the best reply depends on it. Here it does not: confessing is dominant at both of player 2's nodes, so the sequential game unravels exactly as the simultaneous one, and the Pareto-better (N, N) is still out of reach. The mixed Nash equilibria show what subgame perfection removes even here: a "promise" by player 2 to reciprocate silence (N after N) is cheap talk once player 1 has stayed silent, the Lecture 4 lesson that talk is cheap unless something makes it credible (Lecture 4, slide 11). Moving first is a commitment device only when it changes what the follower wants to do (Stackelberg, Lecture 3, slide 37).

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique result: path C, C with payoffs $(-5, -5)$), finds the pure Nash equilibria of the 2×4 normal form (only (C, CC)) and of the simultaneous game (only (C, C)), confirms on a grid of w in steps of $\frac{1}{70}$ that $(C, (1-w)CC + wCN)$ is a Nash equilibrium exactly when $w \leq \frac{5}{7}$, and that no Nash equilibrium puts weight on N. Output line: `munoz-garcia-4-3: SPE (C, CC); NE = (C, CC with prob >= 2/7)`. *Comparison with Muñoz-Garcia's answer* (pp. 160–162): it agrees on all three parts, SPNE $= \{(\text{Confess}, \text{Confess}, \text{Confess})\}$. One wording slip there: the third component is described as player 2's response "to player 1 selecting Confess (in Subgame 3)"; Subgame 3 follows Not Confess, so it is the reply to Not Confess. The book does not list the mixed Nash equilibria.

(f) Common mistakes.

- Writing the SPE as the outcome (C, C): player 2's strategy must say what she does after N too, so the SPE is (C, CC).

- Concluding that the sequential game has a different equilibrium because "player 2 can punish". Punishment after N is not a best reply, so it is not part of any SPE.
- Claiming that (C, CC) is the only Nash equilibrium: it is the only *pure* one, but player 2 may mix off the path ($w \leq \frac{5}{7}$).

Answer. (a) Player 2 confesses after Confess ($-5 > -15$) and after Not confess ($0 > -1$). (b) Player 1 confesses ($-5 > -15$); SPE $= (C, (C, C))$, outcome (C, C), payoffs $(-5, -5)$. Nash equilibria: $(C,\ (1-w)CC + wCN)$, $0 \leq w \leq \frac{5}{7}$; only $w = 0$ is subgame perfect. (c) No difference: the simultaneous game's unique equilibrium is also (C, C), since Confess is strictly dominant at every node.

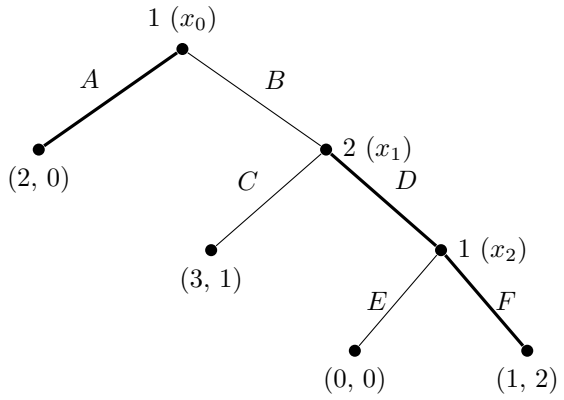
Item 3 (item 23 of the combined list)**: Tadelis Exercise 7.2** Foundations · difficulty 2/5

Source: Tadelis Exercise 7.2, page 147. The statement is reproduced verbatim in the list (item 23); it is not repeated here.

(a) What is asked. For the three-move game A/B, then C/D, then E/F: (a) its tree and type of information; (b) terminal nodes and information sets; (c) pure strategies per player; (d) with the given payoffs, all Nash equilibria and which one is more appealing.

(b) Method. Game tree and information sets (Lecture 3, slides 20–23; Tadelis §7.1, Defs. 7.1–7.2, pp. 133–135); perfect information (slide 24; Def. 7.3, p. 136); pure strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form of an extensive-form game (Tadelis §7.2.3, pp. 142–145); Nash equilibria by best-reply marking and indifference; the "appealing" one by sequential rationality and backward induction (slides 29–30; Tadelis §8.1, pp. 152–153).

(c) Solution. **(a) Tree.** Payoffs (u_1, u_2) from part (d) are written at the leaves.



Thick branches = backward-induction choices (part (d)).

Each decision node is alone in its information set (the mover always knows the history), so it is a game of **perfect information**.

(b) Counts. Terminal nodes: the leaves after A, C, E, F, so **4**. Information sets: $\{x_0\}$, $\{x_2\}$ for player 1 and $\{x_1\}$ for player 2, so **3** (two for player 1, one for player 2).

(c) Pure strategies. Player 1 picks an action at x_0 and at x_2, even when his first action (A) makes x_2 unreachable: $S_1 = \{AE, AF, BE, BF\}$, **4** strategies (2×2, item "Tadelis 7.1"). Player 2 has one information set: $S_2 = \{C, D\}$, **2** strategies.

(d) Nash equilibria. Normal form (rows player 1, columns player 2):

	C	D
AE	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
AF	2, $\underline{0}$	$\underline{2}$, $\underline{0}$
BE	$\underline{3}$, $\underline{1}$	0, 0
BF	$\underline{3}$, 1	1, $\underline{2}$

Pure Nash equilibria (both payoffs underlined): (AE, D), (AF, D) and (BE, C). (BF, C) fails because against BF player 2 prefers D ($2 > 1$).

All Nash equilibria. Let $q = \Pr_2(C)$. Player 1's payoffs are

$$u_1(AE) = u_1(AF) = 2, \qquad u_1(BE) = 3q, \qquad u_1(BF) = 3q + (1-q) = 1 + 2q.$$

- $q < \frac{1}{2}$: $2 > 1 + 2q > 3q$, so player 1 plays A (any mix of AE, AF); player 2's node is not reached, she is indifferent, and any such q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: AE, AF, BF tie at 2. If BF had positive weight, player 2's node would be reached only after BF, where D is strictly better ($2 > 1$), forcing $q = 0$. So player 1 stays on $\{AE, AF\}$. *Equilibria.*
- $\frac{1}{2} < q < 1$: $1 + 2q > \max\{2, 3q\}$, so BF is the unique best reply, and then $q = 0$. Contradiction.
- $q = 1$: BE and BF tie at $3 > 2$. With $w = \Pr_1(BF)$ (the rest on BE), player 2 gets 1 from C and $2w$ from D, so C is a best reply iff $w \leq \frac{1}{2}$. *Equilibria.*

Hence the set of Nash equilibria is

$$\{(\text{any mix of } AE, AF;\ q) : 0 \leq q \leq \tfrac{1}{2}\} \ \cup\ \{((1-w)\,BE + w\,BF;\ C) : 0 \leq w \leq \tfrac{1}{2}\},$$

with outcome A and payoffs $(2, 0)$ in the first family and outcome (B, C), payoffs $(3, 1)$, in the second.

The appealing one. Backward induction: at x_2 player 1 picks F ($1 > 0$); at x_1 player 2, anticipating F, compares $C \mapsto 1$ with $D \mapsto 2$ and picks D; at x_0 player 1 compares $A \mapsto 2$ with $B \mapsto 1$ and picks A. The backward-induction profile, and the unique SPE, is (AF, D), outcome A, payoffs $(2, 0)$. It is the appealing one because every player's plan is optimal at every node, reached or not. The other two pure equilibria rely on player 1's plan to play E at x_2, which gives him $0 < 1$:

- in (BE, C) the plan E is a *threat* that scares player 2 into C; if x_2 were reached, player 1 would not carry it out;
- in (AE, D) the plan E is never tested and does not change the outcome, but it is still not sequentially rational.

(d) Interpretation. Player 1 would like to commit to E ("if you play D I will burn both our payoffs"), because that threat earns him 3 instead of 2. He cannot: at x_2 the damage is in the past and F is better. Nash equilibrium accepts the threat because x_2 is off the path; sequential rationality rejects it. This is the Brexit pattern of Lecture 4 (slides 8–13): several Nash equilibria are held up by threats nobody would execute, and only the backward-induction profile survives.

(e) Verification. `check_B_L03.py` runs backward induction on the tree (unique solution, outcome A, payoffs $(2, 0)$; continuation D then F after B), finds the three pure Nash equilibria of the 4×2 matrix, scans q on a grid of step $\frac{1}{120}$ plus $\frac{1}{2}, \frac{2}{3}$ (Nash equilibria exist exactly for $q \leq \frac{1}{2}$ and $q = 1$), and tests the $w \leq \frac{1}{2}$ condition on a grid. Output line: `tadelis-7-2: SPE (AF, D); pure NE (AE,D), (AF,D), (BE,C)`. Tadelis prints no answer.

(f) Common mistakes.

- Giving player 1 only three strategies (A, BE, BF). That is the reduced normal form; as Tadelis defines strategies, A must be completed with a plan at x_2.
- Calling (BE, C) the "best" equilibrium because it gives player 1 more. A Nash equilibrium is appealing when its threats are credible, not when it pays one player more.

- Counting 3 information sets as "3 decision nodes per player" or counting terminal nodes as information sets.

Answer. (a) Tree above; perfect information. (b) 4 terminal nodes; 3 information sets (2 for player 1, 1 for player 2). (c) Player 1: 4 (AE, AF, BE, BF); player 2: 2 (C, D). (d) Pure Nash equilibria (AE, D), (AF, D), (BE, C) (plus mixed ones: player 1 on A with $\Pr(C) \leq \frac{1}{2}$, or $(1-w)BE + wBF$ with $w \leq \frac{1}{2}$ against C). The appealing one is (AF, D), the unique SPE: the others rest on the non-credible plan E.

Item 4 (item 24 of the combined list)**: Tadelis Exercise 8.5** Exam · difficulty 3/5

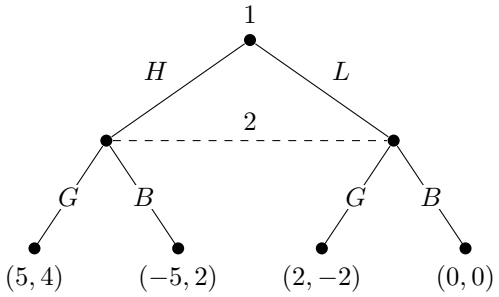
Source: Tadelis Exercise 8.5, page 170. The statement is reproduced verbatim in the list (item 24); it is not repeated here.

(a) What is asked. (a) Tree, Nash and subgame-perfect equilibria of the simultaneous task/training game; (b) tree and number of proper subgames once the CEO first decides whether to adopt (outside payoff $(1, 1)$); (c) Nash and subgame-perfect equilibria of that larger game.

(b) Method. Simultaneous moves as an information set (Lecture 3, slides 22–23; Tadelis Def. 7.2, p. 135); mixed equilibria by the principle of indifference (Lecture 3, slides 10–12); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); SPE by replacing the subgame with each of its Nash equilibria and optimizing at the root (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157). The structure is the voluntary battle of the sexes (Lecture 3, slide 36; Lecture 4, slide 19; Tadelis pp. 154–159).

(c) Solution. Notation. CEO = player 1 (H or L), manager = player 2 (G or B); in (b) the CEO first chooses N (no adoption) or A (adopt). $p = \Pr_1(H)$, $q = \Pr_2(G)$ in the simultaneous game.

(a) Tree and equilibria of the simultaneous game.



Pure: best replies $BR_1(G) = H$ $(5 > 2)$, $BR_1(B) = L$ $(0 > -5)$, $BR_2(H) = G$ $(4 > 2)$, $BR_2(L) = B$ $(0 > -2)$. Pure equilibria: (H, G) with $(5, 4)$ and (L, B) with $(0, 0)$.

Mixed: the manager is indifferent when

$$u_2(G) = 4p - 2(1-p) \;=\; 2p + 0 \cdot (1-p) = u_2(B) \;\Rightarrow\; 6p - 2 = 2p \;\Rightarrow\; p = \tfrac{1}{2}, \tag{1}$$

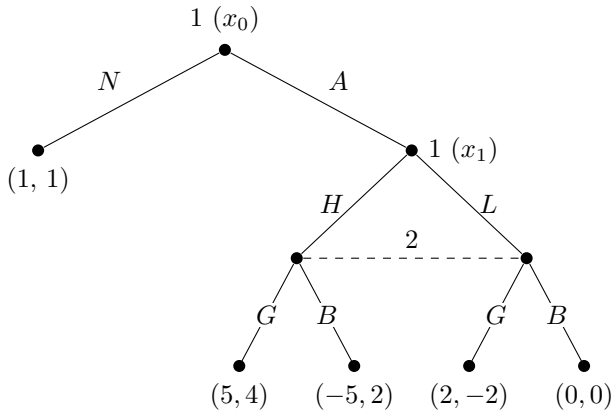
and the CEO is indifferent when

$$u_1(H) = 5q - 5(1-q) \;=\; 2q + 0 \cdot (1-q) = u_1(L) \;\Rightarrow\; 10q - 5 = 2q \;\Rightarrow\; q = \tfrac{5}{8}. \tag{2}$$

Payoffs in the mixed equilibrium: $u_1 = 2q = \frac{5}{4}$, $u_2 = 2p = 1$.

Subgame perfection: the only proper subgame is the game itself (player 2's nodes share an information set, so no subgame starts at them). Every Nash equilibrium is therefore subgame perfect: (H, G), (L, B) and $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$.

(b) The entire game. The CEO moves first at x_0: N ends the game with $(1, 1)$; A leads to x_1, where the game of (a) is played.



Candidate roots of proper subgames are singleton information sets: x_0 and x_1. Player 2's nodes are excluded (each would cut her information set). So there are **two proper subgames**: the whole game and the adoption subgame starting at x_1 (one, if the whole game is not counted, as in Gibbons).

(c) Equilibria of the entire game. The CEO now has two information sets. His strategies are $S_1 = \{NH, NL, AH, AL\}$; the manager's are $S_2 = \{G, B\}$.

	G	B
NH	1, $\underline{1}$	$\underline{1}$, $\underline{1}$
NL	1, $\underline{1}$	$\underline{1}$, $\underline{1}$
AH	$\underline{5}$, $\underline{4}$	-5, 2
AL	2, -2	0, $\underline{0}$

Pure Nash equilibria: (AH, G), (NH, B), (NL, B).

All Nash equilibria. Given q, the CEO's payoffs are 1 (NH, NL), $5q - 5(1-q) = 10q - 5$ (AH) and $2q$ (AL). The manager's gain from G over B is 0 against $N\cdot$, $+2$ against AH and -2 against AL.

- $q < \frac{1}{2}$: $1 > 2q$ and $1 > 10q - 5$, so the CEO plays N (any mix of NH, NL); the manager is never reached and any q is a best reply. *Equilibria.*
- $q = \frac{1}{2}$: N and AL tie at 1 (AH gives 0). For $0 < q < 1$ the manager must be indifferent, i.e. $-2\Pr(AL) = 0$: the CEO stays on N. *Equilibria.*
- $\frac{1}{2} < q < \frac{5}{8}$: AL is the unique best reply ($2q > 1$, $2q > 10q - 5$); then the manager strictly prefers B, $q = 0$. Contradiction.
- $q = \frac{5}{8}$: AH and AL tie at $\frac{5}{4} > 1$; the manager is indifferent iff $2\Pr(AH) - 2\Pr(AL) = 0$, i.e. $\frac{1}{2}AH + \frac{1}{2}AL$. *Equilibrium.*
- $\frac{5}{8} < q \leq 1$: AH is the unique best reply; then the manager plays G, $q = 1$. *Equilibrium* (AH, G) at $q = 1$ only.

So the Nash equilibria are

$$\{(\text{any mix of } NH, NL;\ q) : q \leq \tfrac{1}{2}\} \cup \{(\tfrac{1}{2}AH + \tfrac{1}{2}AL;\ \tfrac{5}{8})\} \cup \{(AH;\ G)\}.$$

Subgame-perfect equilibria. In the adoption subgame play must be one of its three Nash equilibria; replace x_1 by its value and optimize at x_0:

Subgame equilibrium	Value (u_1, u_2)	CEO at x_0	SPE
(H, G)	$(5, 4)$	A $(5 > 1)$	(AH, G), payoffs $(5, 4)$
(L, B)	$(0, 0)$	N $(1 > 0)$	(NL, B), payoffs $(1, 1)$
$(\frac{1}{2}, \frac{5}{8})$	$(\frac{5}{4}, 1)$	A $(\frac{5}{4} > 1)$	$(A,\ \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$, payoffs $(\frac{5}{4}, 1)$

Three SPE. The Nash equilibria that are not subgame perfect are (NH, B) and every profile of the first family other than (NL, B): after adoption they prescribe (H, B) or (L, q) with $q > 0$, or a mix of H and L that is not $\frac{1}{2}$–$\frac{1}{2}$ with $q \leq \frac{1}{2}$, none of which is a Nash equilibrium of the subgame. For instance, in (NH, B) the CEO's plan H against B gives him $-5 < 0$.

(d) Interpretation. Whether a new technology is adopted depends on what each side expects the other to do after adoption, not only on its potential $(5, 4)$. If the CEO expects the manager to train badly he does not adopt, and the expectation is self-fulfilling. Adoption has an option value only if the subgame is expected to coordinate on (H, G), or at least on the mixed play, which gives $\frac{5}{4} > 1$. The outside option $(1, 1)$ works like the UK's no-deal option in the Brexit case (Lecture 4, slide 11): it sets the bar that the continuation must clear, and it eliminates the (H, B) plan, which no rational CEO would carry out after adopting.

(e) Verification. `check_B_L03.py` finds the pure equilibria of the subgame, solves (1)–(2) with sympy, checks the mixed profile and its payoffs $(\frac{5}{4}, 1)$ exactly, finds the pure equilibria (NH, B), (NL, B), (AH, G) of the 4×2 matrix, scans q on a grid of step $\frac{1}{160}$ plus $\frac{1}{2}, \frac{3}{5}, \frac{5}{8}$ (equilibria exactly at $q \leq \frac{1}{2}$, $q = \frac{5}{8}$, $q = 1$), and replaces the subgame by each equilibrium value to get the three SPE. Output line: `tadelis-8-5: subgame NE (H,G), (L,B), (1/2, 5/8); three SPE`. Tadelis prints no answer.

(f) Common mistakes.

- Treating (NH, B) and (NL, B) as the same equilibrium: they are different strategies, and only (NL, B) is subgame perfect.
- Forgetting the mixed equilibrium of the subgame, and with it the third SPE (adoption with mixed play).
- Saying the game of (a) has subgames at player 2's nodes; they belong to one information set.

Answer. (a) Nash (all subgame perfect, since the only subgame is the game itself): (H, G), (L, B), $(\frac{1}{2}H + \frac{1}{2}L,\ \frac{5}{8}G + \frac{3}{8}B)$ with payoffs $(5, 4)$, $(0, 0)$, $(\frac{5}{4}, 1)$. (b) Tree above; two proper subgames (the whole game and the post-adoption game). (c) Pure Nash: (AH, G), (NH, B), (NL, B); all Nash: N (any mix of NH, NL) with $\Pr(G) \leq \frac{1}{2}$, $(\frac{1}{2}AH + \frac{1}{2}AL, \frac{5}{8})$, (AH, G). SPE: (AH, G), (NL, B) and $(A, \frac{1}{2}H + \frac{1}{2}L;\ \frac{5}{8}G)$.

Item 5 (item 25 of the combined list)**: Tadelis Exercise 8.10** Exam · difficulty 3/5

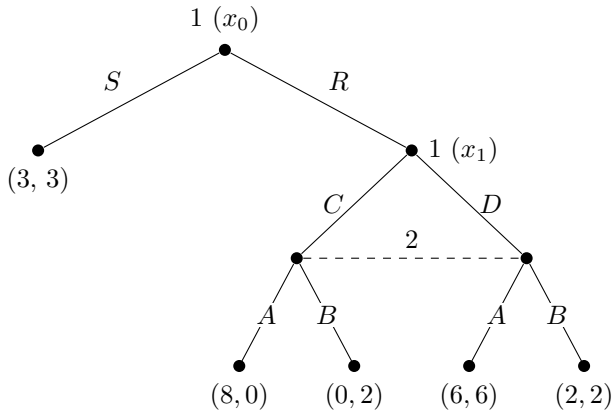
Source: Tadelis Exercise 8.10, page 172. The statement is reproduced verbatim in the list (item 25); it is not repeated here.

(a) What is asked. Player 1 either plays safe (S, payoffs $(3,3)$) or risks (R) a simultaneous 2×2 game with player 2. (a) Draw a tree and count its proper subgames; (b) say whether other trees would do; (c) write the normal form; (d) find all Nash and subgame-perfect equilibria.

(b) Method. Extensive form with an information set for the simultaneous stage (Lecture 3, slides 22–23; Tadelis §7.1, Def. 7.2, p. 135); proper subgames (Lecture 4, slides 14–16; Tadelis Def. 8.2, p. 154, which counts the whole game); strategies as complete plans (Lecture 3, slide 25; Def. 7.4, p. 138); SPE by replacing the subgame with each of its Nash equilibria (Lecture 4, slide 17; Tadelis §8.2, Def. 8.3, p. 157; the voluntary battle of the sexes, Lecture 4, slide 19, has the same structure). Mixed equilibria by the principle of indifference (Lecture 3, slides 10–12; Tadelis §6.1–6.2, pp. 101–114).

(c) Solution. Notation. After R, player 1 chooses C or D and player 2 chooses A or B without seeing it. $p = \Pr_1(C)$, $q = \Pr_2(A)$ in the subgame.

(a) Tree and subgames.



Player 2's two nodes form one information set (dashed). Candidate roots of a proper subgame are the singleton nodes x_0 and x_1; player 2's nodes fail (each would cut her information set). So the game has **two proper subgames**: the whole game and the simultaneous game that starts at x_1 (one, if, like Gibbons, one does not count the whole game).

(b) Other trees. Yes. (i) In the simultaneous stage, player 2 can be drawn as moving first at x_1 with player 1's two nodes joined in an information set; the strategies, normal form and subgames are unchanged. (ii) A different tree also represents the story: player 1 picks S, RC or RD at the root, and player 2 has one information set over the nodes after RC and RD. Its normal form is the same up to merging SC and SD (it is the *reduced* normal form), but it has only one proper subgame, the whole game, so subgame perfection there has no bite: every Nash equilibrium is subgame perfect. The choice of tree matters for SPE whenever it changes which subgames exist; tree (i) and the drawing in (a) are the faithful ones, since R is a real decision taken before the stage game.

(c) Normal form. Player 1 has two information sets, so $S_1 = \{SC, SD, RC, RD\}$ (first letter at x_0, second at x_1); $S_2 = \{A, B\}$:

	A	B
SC	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
SD	3, $\underline{3}$	$\underline{3}$, $\underline{3}$
RC	$\underline{8}$, 0	0, $\underline{2}$
RD	6, $\underline{6}$	2, 2

(d) Equilibria.

Step 1: the subgame at x_1. Best replies: $BR_1(A) = C$ ($8 > 6$), $BR_1(B) = D$ ($2 > 0$), $BR_2(C) = B$ ($2 > 0$), $BR_2(D) = A$ ($6 > 2$). They cycle, so there is no pure equilibrium. Player 2 is indifferent when

$$u_2(A) = 0 \cdot p + 6(1-p) \;=\; 2p + 2(1-p) = u_2(B) \;\Rightarrow\; 6 - 6p = 2 \;\Rightarrow\; p = \tfrac{2}{3}, \tag{3}$$

and player 1 is indifferent when

$$u_1(C) = 8q \;=\; 6q + 2(1-q) = u_1(D) \;\Rightarrow\; 8q = 2 + 4q \;\Rightarrow\; q = \tfrac{1}{2}. \tag{4}$$

The unique equilibrium of the subgame is $(p, q) = (\frac{2}{3}, \frac{1}{2})$, with values

$$u_1 = 8\left(\tfrac{1}{2}\right) = 4, \qquad u_2 = 6\left(1 - \tfrac{2}{3}\right) = 2.$$

Step 2: subgame-perfect equilibrium. Replace x_1 by $(4, 2)$. At x_0, R gives $4 > 3$, so player 1 risks. Since the subgame has a unique equilibrium, the SPE is unique:

$$\sigma^{\text{SPE}}: \text{ player 1 plays } R \text{ and then } C \text{ w.p. } \tfrac{2}{3};\ \text{player 2 plays } A \text{ w.p. } \tfrac{1}{2}, \qquad \text{payoffs } (4, 2).$$

In the normal form this is player 1 mixing $\frac{2}{3}RC + \frac{1}{3}RD$ against $\frac{1}{2}A + \frac{1}{2}B$. There is *no* pure-strategy SPE.

Step 3: all Nash equilibria. Expected payoffs against q:

Player 1's strategy	u_1 given q	player 2's gain $u_2(A) - u_2(B)$ against it
SC, SD	3	0
RC	$8q$	$0 - 2 = -2$
RD	$6q + 2(1-q) = 2 + 4q$	$6 - 2 = +4$

- $q \le \frac{1}{4}$: $3 \ge 2 + 4q$ and $3 > 8q$, so S is a best reply (strictly, for $q < \frac{1}{4}$). With player 1 on $\{SC, SD\}$ in any mix, player 2 is indifferent, so any such q is a best reply. *Nash equilibria.* (At $q = \frac{1}{4}$, RD also ties at 3, but any weight on it would make A strictly better, forcing $q = 1$; so player 1 stays on S.)
- $\frac{1}{4} < q < \frac{1}{2}$: RD is the unique best reply ($2 + 4q > 3$ and $2 + 4q > 8q$); then player 2 strictly prefers A, i.e. $q = 1$. Contradiction.
- $q = \frac{1}{2}$: RC and RD tie at $4 > 3$; player 2 is indifferent only if $-2\Pr(RC) + 4\Pr(RD) = 0$, i.e. $\frac{2}{3}RC + \frac{1}{3}RD$. *The SPE above.*
- $q > \frac{1}{2}$: RC is the unique best reply; then player 2 plays B, $q = 0$. Contradiction.

So the Nash equilibria are

$$\{\,(\text{any mix of } SC, SD;\ q) : 0 \le q \le \tfrac{1}{4}\,\} \;\cup\; \{\,(\tfrac{2}{3}RC + \tfrac{1}{3}RD;\ q = \tfrac{1}{2})\,\},$$

and the *pure* Nash equilibria are (SC, B) and (SD, B), both with payoffs $(3, 3)$. None of the first family is subgame perfect: each asks player 2 to play A with probability at most $\frac{1}{4}$ in a subgame whose only equilibrium has $q = \frac{1}{2}$. The "threat" to play B deters R, but it is not credible once R has been played.

(d) Interpretation. Playing safe is supported only by player 2's threat to play B in the risky game. Once the risky game starts, both must play its (mixed) equilibrium, which gives player 1 an expected 4, more than the safe 3, so he takes the risk. Player 2 is worse off ($2 < 3$): she would like to commit to B, but cannot; a public, costly commitment (Lecture 4, slide 11: observable, sunk moves) would be her only way to keep player 1 on the safe path. The exercise also shows that subgame perfection can select a *mixed* continuation, so the credibility test applies to probabilities, not only to pure threats.

(e) Verification. `check_B_L03.py` confirms that the subgame has no pure equilibrium, solves (3)–(4) with sympy, finds the pure Nash equilibria (SC, B), (SD, B) of the 4×2 matrix, and scans q over a grid of 241 points plus the critical values $\frac{1}{4}, \frac{3}{8}, \frac{1}{2}$: a Nash equilibrium exists exactly at $q \in [0, \frac{1}{4}]$ and $q = \frac{1}{2}$, and at $q = \frac{1}{2}$ the mix $(\frac{2}{3}, \frac{1}{3})$ on (RC, RD) passes the exact Nash test while $(\frac{1}{2}, \frac{1}{2})$ fails. Output line: `tadelis-8-10: unique SPE (R; 2/3 C; 1/2 A), payoffs (4, 2).` Tadelis prints no answer.

(f) Common mistakes.

- Looking for a pure SPE: the subgame has none, so the SPE must use its mixed equilibrium.
- Swapping the indifference conditions: p (player 1's mix) is pinned down by *player 2's* indifference, and vice versa.
- Counting a subgame at each of player 2's nodes, or forgetting that (SC, B) and (SD, B) are two distinct Nash equilibria.

Answer. (a) Tree above; two proper subgames (the whole game and the simultaneous game after R). (b) Yes: player 2 can be drawn first in the stage game; a tree with root choices $\{S, RC, RD\}$ gives the same reduced normal form but no proper subgame other than the whole game. (c) Matrix above, $S_1 = \{SC, SD, RC, RD\}$. (d) Pure Nash: (SC, B), (SD, B); all Nash: player 1 on S (any mix of SC, SD) with $\Pr(A) \leq \frac{1}{4}$, plus $(\frac{2}{3}RC + \frac{1}{3}RD,\ \frac{1}{2}A + \frac{1}{2}B)$. Unique SPE: R, then C w.p. $\frac{2}{3}$; A w.p. $\frac{1}{2}$; payoffs $(4, 2)$. No pure SPE.

Item 6 (item 26 of the combined list)**: Tadelis Exercise 7.4** Exam · difficulty 3/5

Source: Tadelis Exercise 7.4, page 148. The statement is reproduced verbatim in the list (item 26); it is not repeated here.

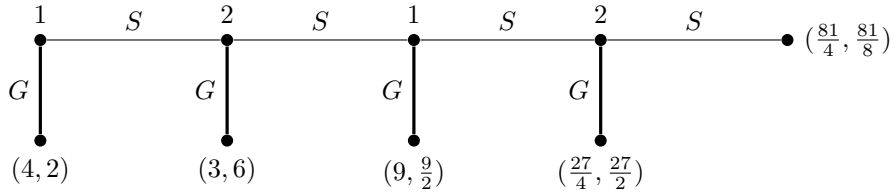
(a) What is asked. In the four-move grab/share game with a pot of \$6 growing by $\frac{3}{2}$: (a) tree and type of information; (b) terminal nodes and information sets; (c) pure strategies; (d) Nash equilibria and how many outcomes they support; (e) the same when sharing at the last node splits the pot equally.

(b) Method. Tree, information sets and perfect information (Lecture 3, slides 20–24; Tadelis §7.1, pp. 133–136); strategies as complete plans (slide 25; Def. 7.4, p. 138); the normal form and its Nash equilibria (Tadelis §7.2.3, pp. 142–145); backward induction to identify the sequentially rational one (slides 29–30; Tadelis §8.1, pp. 152–153; the centipede game itself is Tadelis §8.3.1, pp. 159–160). Nash equilibrium constrains play only on the path, so the outcome argument proceeds forward from the last node that is reached with positive probability.

(c) Solution. Pot and payoffs. The pot before move $k = 1, 2, 3, 4$ is $6 \cdot (\frac{3}{2})^{k-1}$: 6, 9, $\frac{27}{2}$, $\frac{81}{4}$; after a share at move 4 it is $\frac{243}{8}$. A grab gives the mover $\frac{2}{3}$ of the current pot and the other player $\frac{1}{3}$. Payoffs (u_1, u_2):

Terminal history	pot	u_1	u_2
G (1 grabs at move 1)	6	4	2
SG (2 grabs at move 2)	9	3	6
SSG (1 grabs at move 3)	$\frac{27}{2}$	9	$\frac{9}{2}$
$SSSG$ (2 grabs at move 4)	$\frac{81}{4}$	$\frac{27}{4}$	$\frac{27}{2}$
$SSSS$ (2 shares at move 4)	$\frac{243}{8}$	$\frac{81}{4}$	$\frac{81}{8}$

(a) Tree. Perfect information: every information set is a single node.



Thick branches = backward-induction choices (part (d)).

(b) Counts. Terminal nodes: **5** (four grabs and the final share). Information sets: **4**, two per player (moves 1 and 3 for player 1, moves 2 and 4 for player 2).

(c) Pure strategies. Each player has 2 information sets with 2 actions: $2^2 = 4$ pure strategies each. Write $s_1 = (\text{move } 1, \text{move } 3)$ and $s_2 = (\text{move } 2, \text{move } 4)$: $S_1 = S_2 = \{GG, GS, SG, SS\}$.

(d) Nash equilibria. *Backward induction first.* Move 4: player 2 grabs ($\frac{27}{2} > \frac{81}{8}$). Move 3: player 1 grabs ($9 > \frac{27}{4}$). Move 2: player 2 grabs ($6 > \frac{9}{2}$). Move 1: player 1 grabs ($4 > 3$). SPE $= (GG, GG)$, payoffs $(4, 2)$.

Normal form (rows player 1, columns player 2):

	GG	GS	SG	SS
GG	$\underline{4},\ \underline{2}$	$\underline{4},\ \underline{2}$	$4,\ \underline{2}$	$4,\ \underline{2}$
GS	$\underline{4},\ \underline{2}$	$\underline{4},\ \underline{2}$	$4,\ \underline{2}$	$4,\ \underline{2}$
SG	$3,\ \underline{6}$	$3,\ \underline{6}$	$\underline{9},\ \underline{\frac{9}{2}}$	$9,\ \frac{9}{2}$
SS	$3,\ 6$	$3,\ 6$	$\frac{27}{4},\ \underline{\frac{27}{2}}$	$\underline{\frac{81}{4}},\ \frac{81}{8}$

(Underlined payoffs are best replies. Outside the top-left block no cell has both payoffs underlined: in row SS player 2's best reply is SG, against which player 1 prefers SG, against which player 2 prefers to grab at move 2.) The **pure Nash equilibria** are the four profiles $\{GG, GS\} \times \{GG, GS\}$: player 1 grabs at once and player 2 plans to grab at move 2; what either plans at the later, unreached node is free. Every one gives the outcome G with $(4, 2)$.

Mixed equilibria: still one outcome. Work forward along a Nash equilibrium, using that every node reached with positive probability must be played optimally given what follows. (i) If move 4 is reached, player 2 grabs there ($\frac{27}{2} > \frac{81}{8}$). (ii) So if move 3 is reached, sharing gives player 1 $\frac{27}{4}$ against 9 from grabbing: he grabs, and move 4 is never reached. (iii) So if move 2 is reached, sharing gives player 2 $\frac{9}{2}$ against 6 from grabbing: she grabs, and move 3 is never reached. (iv) So sharing at move 1 gives player 1 $3 < 4$: he grabs. Hence **every** Nash equilibrium has player 1 grabbing at move 1: **one outcome**, $(4, 2)$. The mixed equilibria differ only off the path. With $g = \Pr_2(G \text{ at move 2})$ and $h = \Pr_2(G \text{ at move 4})$, player 1's best deviation (share, then the better of grabbing for 9 or sharing for $M = \max\{9,\ \frac{27}{4}h + \frac{81}{4}(1-h)\}$) must not beat 4:

$$3g + M(1-g) \le 4 \iff g \ge 1 - \frac{1}{M-3},$$

e.g. $g \ge \frac{5}{6}$ if $h = 1$ and $g \ge \frac{65}{69}$ if $h = 0$.

(e) Equal split at the end. Sharing at move 4 now gives each player $\frac{1}{2} \cdot \frac{243}{8} = \frac{243}{16} \approx 15.19$. Backward induction: move 4: share ($\frac{243}{16} > \frac{27}{2} = \frac{216}{16}$); move 3: share ($\frac{243}{16} > 9$); move 2: share ($\frac{243}{16} > 6$); move 1: share ($\frac{243}{16} > 4$). SPE $= (SS, SS)$ with payoffs $(\frac{243}{16}, \frac{243}{16})$. The answer to (d) changes: the pure Nash equilibria are now (SS, SS) and, as before, the four profiles $\{GG, GS\} \times \{GG, GS\}$ (player 2's plan to grab at move 2 still deters player 1, and it is never tested). So **two outcomes** are supported: grab at once, $(4, 2)$, and share to the end, $(\frac{243}{16}, \frac{243}{16})$. Grabbing at move 2, 3 or 4 is never an equilibrium outcome: the grabber at move 4 would rather share, the grabber at move 3 would be pre-empted at move 2 (player 2 compares 6 with $\frac{9}{2}$), and a grab at move 2 leaves player 1 with $3 < 4$.

(d) Interpretation. With the original split the mover at the last node always grabs, and the anticipation unravels cooperation back to the first move: the pot that could have grown to \$30 is grabbed at \$6. Changing only the *last* node's rule (an equal split) turns the last mover into someone who wants to continue, and the whole chain of reasoning reverses. Credibility at the end of the game drives behavior at the start, the backward-induction lesson of Lecture 3 and the reason why commitments that change the final payoffs (Lecture 4, slide 11) are so powerful.

(e) Verification. `check_B_L03.py` builds the tree with exact fractions (pots $6, 9, \frac{27}{2}, \frac{81}{4}, \frac{243}{8}$), runs backward induction (unique solution: grab, $(4, 2)$), builds the 4×4 normal form and finds exactly the four pure Nash equilibria, all with outcome $(4, 2)$; it checks the bound on g at $h \in \{0, \frac{1}{2}, 1\}$ on both sides ($\frac{5}{6}$, $\frac{65}{69}$); for (e) it finds the SPE all-share with $(\frac{243}{16}, \frac{243}{16})$ and the five pure Nash equilibria with two outcomes. Output line: `tadelis-7-4: (d) 4 pure NE, one outcome (4, 2); (e) SPE all-share 243/16 each; NE outcomes 2`. Tadelis prints no answer.

(f) Common mistakes.

- Answering "one Nash equilibrium" in (d): there are four pure ones (and a continuum of mixed ones); it is the *outcome* that is unique.
- Misreading the last-node rule in (d): if player 2 shares at move 4 she gets only $\frac{1}{3}$ of the pot, $\frac{81}{8} < \frac{27}{2}$, so she grabs.
- In (e), concluding that sharing is the only equilibrium outcome: the immediate grab is still a (non-subgame-perfect) Nash equilibrium.

Answer. (a) Tree above; perfect information. (b) 5 terminal nodes; 4 information sets (2 each). (c) 4 pure strategies each: GG, GS, SG, SS. (d) Pure Nash equilibria: $\{GG, GS\} \times \{GG, GS\}$; every Nash equilibrium (mixed ones too) has player 1 grabbing at once: one outcome, $(4, 2)$; SPE (GG, GG). (e) Yes: SPE (SS, SS) with $(\frac{243}{16}, \frac{243}{16})$; pure Nash equilibria (SS, SS) and $\{GG, GS\} \times \{GG, GS\}$; two outcomes.

Item 7 (item 27 of the combined list)**: Tadelis Exercise 7.9** Exam · difficulty 4/5

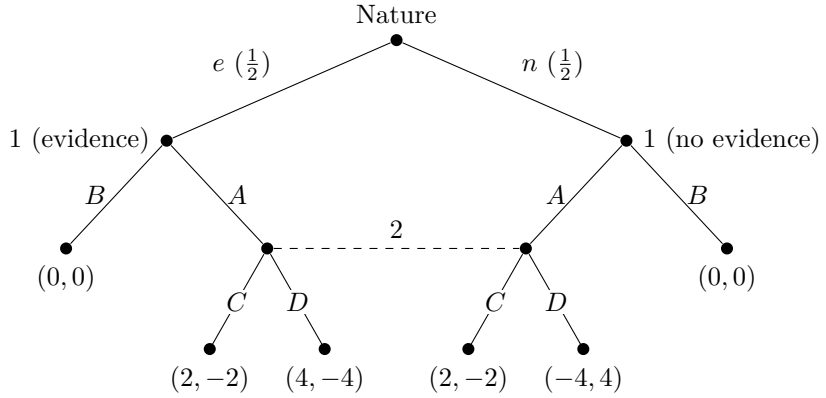
Source: Tadelis Exercise 7.9, page 149–150. The statement is reproduced verbatim in the list (item 27); it is not repeated here.

(a) What is asked. The dean learns whether he has evidence (probability $\frac{1}{2}$), then accuses (A) or bounces (B); an accused student, who does not know whether there is evidence, confesses (C) or denies (D). (a) Game tree; (b) normal-form matrix; (c) Nash equilibria.

(b) Method. A move of Nature at the root and information sets for what a player does not observe (Lecture 3, slides 21–23; Tadelis §7.1, Def. 7.2, p. 135; Nature as in the Brexit tree, Lecture 4, slides 5–6). Strategies as plans over information sets (Lecture 3, slide 25), with expected payoffs over Nature's move in the normal form (Tadelis §7.2.3, pp. 142–145). Nash equilibria by dominance, best replies and the principle of indifference (Lecture 3, slides 10–12).

(c) Solution. Notation. Nature draws e (evidence) or n (no evidence), each with probability $\frac{1}{2}$. The dean (player 1) observes it. Payoffs (u_1, u_2) = (dean, student): $B \mapsto (0,0)$; $A, C \mapsto (2,-2)$; A, D with evidence $\mapsto (4,-4)$; A, D without evidence $\mapsto (-4,4)$. Note that $u_1 + u_2 = 0$ at every leaf: the game is zero-sum.

(a) Tree. The student's two nodes (after eA and after nA) form one information set: he knows he was accused, not whether there is evidence.



(b) Normal form. The dean has two information sets, so a strategy is a pair (action with evidence, action without): $S_1 = \{AA, AB, BA, BB\}$; $S_2 = \{C, D\}$. Each cell is $\frac{1}{2}\times$(payoff after e) $+\frac{1}{2}\times$(payoff after n). For example, $u(AB, D) = \frac{1}{2}(4,-4) + \frac{1}{2}(0,0) = (2,-2)$ and $u(AA, D) = \frac{1}{2}(4,-4) + \frac{1}{2}(-4,4) = (0,0)$.

	C	D
AA	$\underline{2}$, -2	0, $\underline{0}$
AB	1, $\underline{-1}$	$\underline{2}$, -2
BA	1, -1	-2, $\underline{2}$
BB	0, $\underline{0}$	0, $\underline{0}$

(c) Nash equilibria.

Step 1: no pure equilibrium. No cell has both payoffs underlined: against C the dean plays AA, against which the student denies; against D the dean plays AB, against which the student confesses. The best replies cycle.

Step 2: dominated strategies. BA is strictly dominated by AA ($2 > 1$, $0 > -2$): accusing only when there is no evidence is never sensible. BB is strictly dominated by AB ($1 > 0$, $2 > 0$). So in any equilibrium the dean accuses whenever he has evidence and mixes only between AA and AB, i.e. over what he does *without* evidence.

Step 3: indifference. Let $q = \Pr_2(C)$ and $p = \Pr_1(AA)$ (with $1-p$ on AB). The dean is indifferent when

$$u_1(AA) = 2q \;=\; q + 2(1-q) = u_1(AB) \;\Rightarrow\; 2q = 2 - q \;\Rightarrow\; q = \tfrac{2}{3}, \tag{5}$$

and the student is indifferent when

$$u_2(C) = -2p - (1-p) \;=\; 0 \cdot p - 2(1-p) = u_2(D) \;\Rightarrow\; -1 - p = -2 + 2p \;\Rightarrow\; p = \tfrac{1}{3}. \tag{6}$$

At $q = \frac{2}{3}$ the dean's payoffs are $AA, AB : \frac{4}{3}$, $BA : 3q - 2 = 0$, $BB : 0$, so mixing over AA and AB is optimal. The unique Nash equilibrium is

$$\sigma_1^* = \tfrac{1}{3}\,AA + \tfrac{2}{3}\,AB, \qquad \sigma_2^* = \tfrac{2}{3}\,C + \tfrac{1}{3}\,D, \qquad (u_1, u_2) = (\tfrac{4}{3},\ -\tfrac{4}{3}).$$

Uniqueness. For $q < \frac{2}{3}$ only AB is a best reply and the student would then confess ($-1 > -2$), so $q = 1$; for $q > \frac{2}{3}$ only AA is a best reply and the student would deny ($0 > -2$), so $q = 0$; at $q = \frac{2}{3}$ only $p = \frac{1}{3}$ keeps the student indifferent.

Behavioral reading. With evidence the dean always accuses; without evidence he accuses (bluffs) with probability $\frac{1}{3}$. An accused student therefore faces evidence with probability $\frac{1/2}{1/2+1/2\cdot 1/3} = \frac{3}{4}$, and at that belief C and D both give -2 (the check: $\frac{3}{4}(-4) + \frac{1}{4}(4) = -2$).

(d) Interpretation. The dean must bluff sometimes: if he accused only with evidence, every accusation would reveal evidence and the student would confess, which in turn makes accusing without evidence profitable. The equilibrium bluffing rate keeps the student exactly indifferent. Because the game is zero-sum, $\frac{4}{3}$ is its value: the dean cannot guarantee more, the student cannot hold him to less. The structure (Nature, a privately informed first mover, a responder who sees only the action) is the one Lecture 4 used for the Brexit negotiation, and it is the bridge to games of incomplete information later in the course.

(e) Verification. `check_B_L03.py` builds the matrix from the tree with exact fractions (it equals the one above), checks the zero-sum property and that there is no pure equilibrium, scans q on a grid of step $\frac{1}{300}$ (an equilibrium exists only at $q = \frac{2}{3}$), verifies $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$ with the exact Nash test, and solves (6) with sympy. Output line: `tadelis-7-9: unique NE dean 1/3 AA + 2/3 AB, student C w.p. 2/3; value 4/3`. Tadelis prints no answer.

(f) Common mistakes.

- Giving the dean only two strategies (A, B): he chooses after observing Nature, so a strategy is a pair of actions.
- Drawing the student's two nodes as separate information sets: then he would know whether there is evidence, and the game would change completely.
- Using payoffs conditional on one state in the matrix instead of the $\frac{1}{2}$–$\frac{1}{2}$ expectation.

Answer. (a) Tree above, with the student's nodes in one information set. (b) Matrix above, rows AA, AB, BA, BB (action with evidence, without). (c) No pure equilibrium; unique Nash equilibrium $(\frac{1}{3}AA + \frac{2}{3}AB,\ \frac{2}{3}C + \frac{1}{3}D)$: the dean always accuses with evidence and bluffs w.p. $\frac{1}{3}$ without it; the student confesses w.p. $\frac{2}{3}$; payoffs $(\frac{4}{3}, -\frac{4}{3})$.

Item 8 (item 28 of the combined list)**: Tadelis Exercise 7.10** Exam · difficulty 4/5

Source: Tadelis Exercise 7.10, page 150. The statement is reproduced verbatim in the list (item 28); it is not repeated here.

(a) What is asked. In the game of Figure 7.14: (a) pure-strategy sets; (b) every behavioral strategy of player 1 has an outcome-equivalent mixed strategy; (c) the same for player 2; (d) whether (c) fails once player 2's two bottom information sets are merged.

(b) Method. Pure, mixed and behavioral strategies (Lecture 3, slides 25 and 27; Tadelis §7.2.1–7.2.2, Defs. 7.4–7.6, pp. 137–140); perfect recall and Kuhn's equivalence theorem (Lecture 3, slide 28; Tadelis Def. 7.7 and the discussion on pp. 141–142). Outcome equivalence is checked node by node: two strategies of a player are equivalent if, against every pure strategy of the opponent (and then, by linearity, every mixed one), they put the same probability on each terminal node.

(c) Solution. The game. Player 2 moves first (a or b). After a, player 1 chooses L or R; after b, he chooses W or E. Player 2 then moves again without seeing player 1's choice: at information set $h_a = \{aL, aR\}$ with actions t, z, and at $h_b = \{bW, bE\}$ with actions x, y. Payoffs (u_1, u_2): $(2,1)$ after Lt and Wx, $(1,2)$ after Rz and Ey, $(0,0)$ otherwise.

(a) Pure strategies. Player 1 has two information sets (after a, after b), each with two actions:

$$S_1 = \{L, R\} \times \{W, E\} = \{LW, LE, RW, RE\} \quad (4).$$

Player 2 has three information sets (the root, h_a, h_b):

$$S_2 = \{a, b\} \times \{t, z\} \times \{x, y\} \quad (2^3 = 8),$$

e.g. (a,t,x), (a,t,y), …, (b,z,y).

(b) Player 1: behavioral ⇒ mixed. A behavioral strategy is $\beta_1 = (\lambda, \omega)$, with $\lambda = \Pr(L)$ at the left node and $\omega = \Pr(W)$ at the right node. Define the mixed strategy as the product

$$\sigma_1(LW) = \lambda\omega, \quad \sigma_1(LE) = \lambda(1-\omega), \quad \sigma_1(RW) = (1-\lambda)\omega, \quad \sigma_1(RE) = (1-\lambda)(1-\omega).$$

Fix any pure $s_2 = (r, m_a, m_b)$. If $r = a$, the play reaches player 1's left node and ends at (a, L, m_a) or (a, R, m_a). Under β_1 these have probabilities λ and $1-\lambda$. Under σ_1, (a, L, m_a) occurs iff the drawn plan starts with L:

$$\sigma_1(LW) + \sigma_1(LE) = \lambda\omega + \lambda(1-\omega) = \lambda,$$

and (a, R, m_a) has probability $1-\lambda$. The case $r = b$ is symmetric with ω. All other terminal nodes have probability 0 under both. So β_1 and σ_1 induce the same distribution against every pure s_2, hence (expected values are linear in σ_2) against every mixed σ_2 too. Since player 1 is called once per play, only the marginals matter: any σ_1 with $\sigma_1(LW) = \mu$, $\sigma_1(LE) = \lambda - \mu$, $\sigma_1(RW) = \omega - \mu$, $\sigma_1(RE) = 1 - \lambda - \omega + \mu$, $\mu \in [\max\{0, \lambda+\omega-1\}, \min\{\lambda, \omega\}]$, also works.

(c) Player 2: behavioral ⇒ mixed. A behavioral strategy is $\beta_2 = (\alpha, \tau, \xi)$: $\Pr(a)$ at the root, $\Pr(t)$ at h_a, $\Pr(x)$ at h_b. Define the product mixed strategy

$$\sigma_2(r, m_a, m_b) = \Pr_\alpha(r) \cdot \Pr_\tau(m_a) \cdot \Pr_\xi(m_b),$$

e.g. $\sigma_2(a,t,x) = \alpha\tau\xi$. Fix any pure $s_1 = (\ell, m)$ with $\ell \in \{L, R\}$, $m \in \{W, E\}$. Under β_2, terminal node (a, ℓ, t) has probability $\alpha\tau$. Under σ_2 it occurs iff the drawn plan has $r = a$ and $m_a = t$ (any m_b):

$$\sum_{m_b \in \{x,y\}} \sigma_2(a, t, m_b) = \alpha\tau\xi + \alpha\tau(1-\xi) = \alpha\tau.$$

The same computation gives $\alpha(1-\tau)$ for (a,ℓ,z), $(1-\alpha)\xi$ for (b,m,x) and $(1-\alpha)(1-\xi)$ for (b,m,y); nodes off s_1's choices have probability 0 under both. So the two strategies are outcome-equivalent against every s_1, hence against every σ_1. What makes this work is *perfect recall*: at h_a player 2 remembers she chose a, so her bottom move can be drawn independently of, and after, her root move, and the product of independent draws reproduces it.

(d) Imperfect recall. Merge h_a and h_b into one information set h. Actions at the nodes of an information set must be the same, so identify $x \equiv t$ and $y \equiv z$. Player 2 now has $S_2' = \{a,b\}\times\{t,z\}$ (4 pure strategies); a behavioral strategy is (α,τ); she has *forgotten* her own first move.

The claim of (c) survives. The product mixed strategy $\sigma_2(r,m) = \Pr_\alpha(r)\Pr_\tau(m)$ still reproduces (α,τ): each play passes through h at most once, so the probability of a terminal node (r,ℓ,m) is $\Pr_\alpha(r)\Pr_\tau(m)$ under both. No example can break (c) in this game. (Behavioral $\Rightarrow$ mixed fails only when one play can cross the same information set twice, as in the absent-minded driver, Tadelis p. 142.)

What fails is the converse (mixed $\Rightarrow$ behavioral, the other half of Kuhn's theorem). Take

$$\sigma_2 = \tfrac{1}{2}\,(a,t) + \tfrac{1}{2}\,(b,z)$$

("go left and then play t, or go right and then play z": the plan coordinates the bottom move with the root move she no longer remembers). Against $s_1 = LW$ it gives $\Pr(a,L,t) = \frac{1}{2}$, $\Pr(b,W,z) = \frac{1}{2}$, $\Pr(a,L,z) = 0$. A behavioral (α,τ) gives

$$\Pr(a,L,t) = \alpha\tau, \qquad \Pr(a,L,z) = \alpha(1-\tau), \qquad \Pr(b,W,z) = (1-\alpha)(1-\tau).$$

$\Pr(a,L,z) = 0$ forces $\alpha = 0$ or $\tau = 1$; the first makes $\Pr(a,L,t) = 0$, the second makes $\Pr(b,W,z) = 0$, contradicting $\frac{1}{2}$ in either case. So this mixed strategy has no outcome-equivalent behavioral strategy. It also matters for payoffs: against LW, σ_2 gives player 2 $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 0 = \frac{1}{2}$, and against RE it gives $\frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 2 = 1$; behavioral strategies cannot replicate this coordination.

(d) Interpretation. Mixed strategies are plans chosen before the game; behavioral strategies are choices made at each information set. With perfect recall the two are interchangeable, which is why the course (and Tadelis) can analyze dynamic games with behavioral strategies and define SPE with them (Lecture 3, slide 34). Without recall a player who randomizes "in the moment" cannot correlate her later move with an earlier one she has forgotten, while a plan made in advance can.

(e) Verification. `check_B_L03.py` computes with sympy, for each pure strategy of the opponent and each of the eight terminal nodes, the probability under the behavioral strategy and under the product mixed strategy, for player 1 (b) and player 2 (c); all differences simplify to 0. For (d) it confirms the product construction still works and that the system $\alpha\tau = \frac{1}{2}$, $\alpha(1-\tau) = 0$, $(1-\alpha)(1-\tau) = \frac{1}{2}$ has no solution. Output line: `tadelis-7-10: (b), (c) equivalences hold symbolically; (d) mixed->behavioral fails`. Tadelis prints no answer. **Note on the statement:** read literally, (d) asks for a failure of behavioral $\Rightarrow$ mixed, which cannot fail here (each play crosses the merged set once); the example above shows the failure of the converse, which is the equivalence that perfect recall protects in this game.

(f) Common mistakes.

- Giving player 2 only $2\times 2 = 4$ strategies in (a): with perfect recall she has two separate bottom information sets, so $2^3 = 8$.
- Proving (b) only against one strategy of player 2: equivalence must hold against all of them.

- In (d), claiming that the product construction breaks. It does not; the coordination of a mixed plan is what behavioral strategies lose.

Answer. (a) $S_1 = \{LW, LE, RW, RE\}$; $S_2 = \{a, b\} \times \{t, z\} \times \{x, y\}$ (8 strategies). (b), (c) The product of the behavioral probabilities is a mixed strategy with the same distribution over terminal nodes against every opponent strategy. (d) Behavioral $\Rightarrow$ mixed still holds (each play crosses the merged set once); the equivalence that fails is mixed $\Rightarrow$ behavioral: $\frac{1}{2}(a, t) + \frac{1}{2}(b, z)$ has no behavioral equivalent.

Item 9 (item 29 of the combined list)**: MWG Exercise 9.B.4** Advanced · difficulty 4/5

Source: MWG Exercise 9.B.4, page 302. The statement is reproduced verbatim in the list (item 29); it is not repeated here.

(a) What is asked. Prove that in a finite zero-sum game of perfect information all subgame-perfect Nash equilibria give the same payoffs.

(b) Method. Backward induction and Zermelo's theorem (MWG §9.B, Prop. 9.B.1, p. 272; Lecture 3, slide 30, whose corollary gives uniqueness only when no player has equal payoffs at two terminal nodes); subgames and SPNE (MWG Defs. 9.B.1–9.B.2, pp. 274–275; Lecture 3, slides 32–34); in a finite game of perfect information the SPNE are exactly the backward-induction profiles (MWG Prop. 9.B.2; Lecture 3, slide 35). The proof is an induction on subgames that tracks *values*, not strategies: ties may create many SPNE, and the point is that they cannot create different payoffs.

(c) Solution. Setting. Two players; a finite tree in which every information set is a single node; $u_1(z) + u_2(z) = 0$ at every terminal node z, so we can track $v(z) = u_1(z)$ and $u_2(z) = -v(z)$. For a decision node x, Γ_x is the subgame starting at x (every node starts one, by perfect information), and $C(x)$ is the set of its immediate successors. An SPNE may use behavioral strategies.

Claim. For every node x there is a number $V(x)$ such that *every* SPNE of Γ_x gives player 1 the payoff $V(x)$ (and player 2 $-V(x)$).

Proof by induction on the length of the longest path from x to a terminal node.

Base. If x is terminal, $V(x) = v(x)$.

Step. Let x be a decision node of player i and suppose the claim holds for every $y \in C(x)$. Take any SPNE σ of Γ_x.

(i) For each $y \in C(x)$, the restriction of σ to Γ_y is an SPNE of Γ_y (a subgame of Γ_y is a subgame of Γ_x). By the induction hypothesis it gives player 1 exactly $V(y)$. So if play moves from x to y, the continuation pays $(V(y), -V(y))$, whatever σ prescribes inside Γ_y.

(ii) The restriction of σ to Γ_x is a Nash equilibrium of Γ_x. Player i could change only his action at x, keeping the rest of σ; moving to y would give him $V(y)$ if $i = 1$, or $-V(y)$ if $i = 2$. Therefore every action played with positive probability at x must lead to a maximizer:

$$i = 1: \quad y \in \arg\max_{y' \in C(x)} V(y'), \qquad i = 2: \quad y \in \arg\max_{y' \in C(x)} \big(-V(y')\big) = \arg\min_{y' \in C(x)} V(y').$$

(iii) All the successors in the support therefore have the same value, so the payoff of σ in Γ_x is

$$V(x) = \max_{y \in C(x)} V(y) \ \text{ if } i = 1, \qquad V(x) = \min_{y \in C(x)} V(y) \ \text{ if } i = 2,$$

a number that does not depend on which SPNE σ we took.

Applying the claim at the root: every SPNE of the game gives $(V(x_0), -V(x_0))$. SPNE exist by Zermelo's theorem, so the set of SPNE payoffs is exactly $\{(V(x_0), -V(x_0))\}$. □

Where zero-sum is used. In step (iii). When player i is indifferent among several successors, they give *him* the same payoff, and in a two-player zero-sum game they then give the opponent the same payoff too. Without zero-sum, the opponent's payoff can differ across the tied successors. Counterexample: player 1 chooses $l \mapsto (0, 1)$ or $r \mapsto (0, 2)$; both l and r are SPNE, with different payoffs for player 2.

Two remarks.

- *Two players are needed.* With three players and $u_1 + u_2 + u_3 = 0$ the result fails: player 1 choosing between $(0, 1, -1)$ and $(0, -1, 1)$ has two SPNE with different payoffs for players 2 and 3. "Zero-sum" in the exercise means two-player zero-sum.
- *Chance moves.* If the tree has moves of Nature with known probabilities, define $V(x)$ at a chance node as the expected value of its successors' values; the same induction goes through. By the minimax theorem $V(x_0)$ is also the value of the game, and every Nash equilibrium, not only every SPNE, yields it.

(d) Interpretation. In a pure conflict, ties are harmless: if I am indifferent between two moves, so are you, so my tie-breaking cannot be used to help or hurt you. That is why chess-like games have a well-defined value even though the backward-induction strategies are far from unique. In the non-zero-sum games of Lecture 4 (entry, Brexit) a follower's indifference *is* strategically important: how she breaks it can change the leader's payoff (compare MWG 9.B.5, item "MWG 9.B.5", part (e)).

(e) Verification. `check_B_L03.py` draws 400 random two-player zero-sum trees (depth up to 4, two or three moves per node, integer payoffs in $\{-2, \ldots, 2\}$, fixed seed), computes the set of pure SPNE payoff vectors over all tie-breakings, and finds exactly one in each tree; 151 of the trees have more than one SPNE, so ties do occur. It also confirms the two counterexamples (non-zero-sum and three-player zero-sum) have two SPNE payoff vectors. Output line: `mwg-9-b-4: 400 random zero-sum trees, unique SPNE payoffs (151 with several SPNE)`. *Comparison with the MWG solutions manual* (Exercise 9.B.4, p. 9-2 of the manual; label read as "9.8.4" in the OCR): it argues that if a player has equal payoffs at two terminal nodes, the zero-sum property makes the other player indifferent too, so "there may exist many different ... SPNE ... but all of them will yield the same payoffs". **Agreement** on the result; the manual's argument speaks of terminal nodes only, and the induction above extends it to ties between subgame values, which is where it is actually needed.

(f) Common mistakes.

- Proving that the SPNE is unique: it need not be (ties); only the payoffs are.
- Using Zermelo's theorem's uniqueness clause, which requires no ties at all, instead of an argument that allows ties.
- Forgetting that SPNE may be mixed at tied nodes; the argument must cover any probability on the maximizers, which (ii)–(iii) do.

Answer. By induction on subgames: if every subgame after x has a unique SPNE value $V(y)$ for player 1, then at x any SPNE puts weight only on successors that maximize the mover's payoff, $\max_y V(y)$ for player 1 or $\min_y V(y)$ for player 2, so $V(x)$ is unique. At the root all SPNE pay $(V(x_0), -V(x_0))$. Zero-sum (with two players) is what makes the mover's ties irrelevant for the opponent.

Item 10 (item 30 of the combined list)**: Gibbons Problem 2.1** Advanced · difficulty 5/5

Source: Gibbons Problem 2.1, page 130. The statement is reproduced verbatim in the list (item 30); it is not repeated here.

(a) What is asked. Prove Becker's Rotten Kid theorem: when a child's action A produces incomes $I_C(A)$, $I_P(A)$ and an altruistic parent then chooses a bequest B, the backwards-induction outcome has the child choosing the A that maximizes $I_C(A) + I_P(A)$.

(b) Method. Backward induction in a two-stage game of perfect information with continuous actions (Lecture 3, slides 29–30; Gibbons §2.1.A, pp. 57–61; Tadelis §8.3.2): solve the parent's problem for every pair of incomes, then the child's problem given the parent's reply. The key step is that the parent's reply depends on the incomes only through their sum, and raises the child's consumption when the sum rises.

(c) Solution. Notation and assumptions. $A \geq 0$ the child's action; I_C, I_P strictly concave with maximizers $A_C, A_P > 0$; $B \in \mathbb{R}$ the bequest; U, V increasing and strictly concave; $k > 0$. Write $Y = I_C + I_P$ (family income) and $x = I_C + B$ (the child's consumption), so the parent consumes $I_P - B = Y - x$. As is standard for this problem, U and V are differentiable and the parent's problem has a solution.

Step 1: the parent's problem depends only on Y. Given (I_C, I_P) the parent solves

$$\max_{B \in \mathbb{R}} \; V(I_P - B) + k\,U(I_C + B).$$

Because B is unrestricted in sign, choosing B is the same as choosing $x = I_C + B \in \mathbb{R}$, and the problem becomes

$$\max_{x \in \mathbb{R}} \; W(x; Y) = V(Y - x) + k\,U(x). \tag{7}$$

W is strictly concave in x (a sum of strictly concave functions), so the maximizer $x^*(Y)$ is unique and satisfies the first-order condition

$$V'(Y - x^*) = k\,U'(x^*). \tag{8}$$

The bequest is $B^*(I_C, I_P) = x^*(I_C + I_P) - I_C$: it depends on both incomes, but the child's consumption depends only on their sum.

Step 2: x^* is strictly increasing in Y. Take $Y' > Y$ and suppose $x' = x^*(Y') \leq x = x^*(Y)$. Then $Y' - x' > Y - x$. Strict concavity makes V' strictly decreasing and U' decreasing, so

$$V'(Y' - x') < V'(Y - x) \qquad \text{and} \qquad k\,U'(x') \geq k\,U'(x).$$

Applying (8) at Y and at Y':

$$V'(Y' - x') = k\,U'(x') \;\geq\; k\,U'(x) = V'(Y - x) \;>\; V'(Y' - x'),$$

a contradiction. Hence $x^*(Y') > x^*(Y)$. (With U, V twice differentiable, $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$: the parent passes part, not all, of an income gain to the child.)

Step 3: the child's problem. Anticipating B^*, the child chooses $A \geq 0$ to maximize

$$U\big(I_C(A) + B^*(I_C(A), I_P(A))\big) = U\big(x^*(I_C(A) + I_P(A))\big) = U\big(x^*(Y(A))\big).$$

$U \circ x^*$ is strictly increasing (Steps 1–2 and U increasing), so A maximizes the child's payoff if and only if it maximizes $Y(A) = I_C(A) + I_P(A)$.

Step 4: the maximizer exists and is unique. Y is strictly concave. For $A < \min\{A_C, A_P\}$ both incomes are increasing, and for $A > \max\{A_C, A_P\}$ both are decreasing, so Y attains its maximum on the compact interval $[\min\{A_C, A_P\}, \max\{A_C, A_P\}] \subset (0, \infty)$, at a unique point A^*. The backwards-induction outcome is

$$(A^*,\ B^*(I_C(A^*), I_P(A^*))), \qquad A^* = \arg\max_{A \geq 0} \left[\, I_C(A) + I_P(A) \,\right]. \qquad \square$$

Worked example (exact). $U = V = \ln$, $k = 2$, $I_C = 4A - A^2$ ($A_C = 2$), $I_P = 10A - A^2$ ($A_P = 5$). The parent's condition (8) reads $\frac{1}{Y-x} = \frac{2}{x}$, so $x^*(Y) = \frac{2Y}{3}$ (in general $x^* = \frac{kY}{1+k}$). The child maximizes $\ln(\frac{2}{3}(14A - 2A^2))$: $14 - 4A = 0$, $A^* = \frac{7}{2}$, between $A_C = 2$ and $A_P = 5$. There $Y = \frac{49}{2}$, $x^* = \frac{49}{3}$, $I_C = \frac{7}{4}$, so the bequest is $B^* = \frac{49}{3} - \frac{7}{4} = \frac{175}{12}$. Had the child chosen its own income maximizer $A_C = 2$, family income would be $Y(2) = 20 < \frac{49}{2}$ and its consumption only $\frac{40}{3} < \frac{49}{3}$.

(d) Interpretation. The child is selfish but faces a parent who "taxes" any loss of family income: if the child raised I_C at a larger cost to I_P, the parent, who cares about the split, would cut the bequest by more than the child gained. Altruism on one side, plus the parent's last move, aligns the first mover's incentives with joint surplus. Each assumption has a role:

- *the parent moves last*: it is the parent's credible, sequentially rational reply that disciplines the child (Lecture 4's theme: what matters is what a player will actually do when the moment comes);
- *B unrestricted*: if $B \geq 0$ were imposed and binding, the child's consumption would be I_C alone over a range and Step 1 fails;
- *the child moves first and cannot commit the parent*: if the bequest were fixed before A, the child would simply choose A_C.

(e) Verification. `check_B_L03.py` checks with sympy that $dx^*/dY = V''/(V'' + kU'') \in (0, 1)$ for several negative (V'', U'') and positive k; derives $x^* = kY/(1+k)$ for log utility (second derivative negative) and $x^* = (Y + \ln k)/2$ for CARA utility $-e^{-x}$; solves the child's problem in the log example ($A^* = \frac{7}{2}$), confirms on an exact grid of step $\frac{1}{20}$ that the maximizer of $\frac{2}{3}Y(A)$ is $\frac{7}{2}$ while I_C alone peaks at 2, and recomputes $B^* = \frac{175}{12}$. Output line: `gibbons-2-1: x*(Y) increasing; child picks A* = 7/2 = argmax(I_C + I_P), not A_C = 2`. Gibbons prints no answer.

(f) Common mistakes.

- Assuming the child maximizes I_C: its payoff depends on the bequest, which reacts to A.
- Treating the bequest as a fixed number instead of a function $B^*(I_C, I_P)$ (a strategy of the parent in the second stage).
- Proving only that x^* is weakly increasing, or skipping existence/uniqueness of A^*; the strict monotonicity in Step 2 is what makes "maximize $U(x^*(Y))$" equivalent to "maximize Y".

Answer. The parent's best bequest gives the child consumption $x^*(I_C + I_P)$ that solves $V'(Y - x) = kU'(x)$; it depends only on family income Y and is strictly increasing in it. So the child's payoff $U(x^*(Y(A)))$ is a strictly increasing function of $Y(A)$, and the child chooses $A^* = \arg\max_A [I_C(A) + I_P(A)]$, which exists and is unique by strict concavity.